VARUN SESHU

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EDUCATION

Stevens Institute of Technology — May 2024
Master of Science in Financial Engineering — Hoboken, NJ
Concentration: Algorithmic Trading Strategies — GPA: 3.9/4.0

PES University — May 2022
B.Tech. Computer Science Engineering — Bangalore, India
Specialization: Machine Intelligence and Data Science — CGPA: 8.59/10

SKILLS

Programming languages: Python, C++, JavaScript, Bash, SQL
Electronic Trading & Market Infrastructure: Algorithmic trading workflows, Order lifecycle analysis, FIX protocol
Frameworks & Tools: Pandas, Django, Excel, Vue.js
Strengths: Analytical problem-solving under pressure, clear and effective communication, detail-oriented, quick learner, and collaborative team player

PROFESSIONAL EXPERIENCE

Liquidnet (TP ICAP Group) — September 2025 - Present
Front Office Product Support — New York City, NY
- Supported electronic and algorithmic trading workflows for institutional equity execution, resolving time-sensitive production issues with traders, sales, and engineering.
- Selected to represent the Americas region on Liquidnet's global AI workstream, driving adoption of AI tools and building new ones across regional Product Support teams to reduce duplicated effort and improve consistency.
- Diagnosed order flow, configuration, and connectivity issues affecting algorithmic trading performance
- Managed OMS/EMS configuration changes and incident response, including post-trade analysis to reduce recurring issues.
- Built and deployed an internal deployment tool for Inforeach OMS GUIs, streamlining release workflows and reducing manual steps for the front-office team.

Bestex Research — August 2024 - August 2025
Execution Services Analyst — Stamford, CT
- Certified and managed FIX connectivity for multiple brokers and venues, ensuring reliable trade routing and compliance.
- Supported real-time trading operations, resolving production issues in coordination with traders, engineers, and venues.
- Streamlined onboarding for equities and futures clients, reducing setup time by automating the certification process leveraging FIX protocol expertise.
- Owned and improved the UAT testing environment by automating operational workflows with Python scripts, cutting manual effort.

Hiver — January 2022 – July 2022
Software Engineering Intern — Bangalore, India
- Developed Hiver Chat app’s frontend using Vue.js and ReactJS, opening a new channel for customer support.
- Integrated 8 customization features to Hiver Chat app by adding functionality to Hiver Admin Panel using ReactJS.
- Implemented an interactive analytics dashboard with Vue.js, enabling customers to track chat engagement and customer feedback, resulting in a 5% increase in customers feedback responses.

ACADEMIC PROJECTS

AI Algorithmic Trading System | Python, Django, Vue.js | [Github](), [Research paper]()
- Trained 150 LSTM models leveraging Pandas, Keras and Django to predict prices of stocks in NIFTY 50 index.
- Ran 2000 backtests resulting in 1.2 million (simulated) orders, 600,000 buy/sell signals and 600,000 trades and generated over 2000 paper trades.
- Published an IEEE research paper after analyzing results of backtests.
- Visualized strategy performance, backtests and paper trades by building a website utilizing Vue.js.

American Options Pricing Engine | C++, Quantlib | [Github]()
- Designed and implemented an American call and put options pricing engine using binomial and random tree methods.
- Priced 6 combinations of American options utilizing factory and observer design patterns using C++ and Quantlib.

Portfolio Allocation Models Comparison | Python, Pandas | [Github]()
- Built and compared 2 variations of factor-based long/short portfolio allocation models with constraints on their betas.
- Backtested 90 combinations of models and estimators using computed weekly portfolio rebalancing weights.
- Utilized metrics computed on all backtests of both models over 6 different time periods to compare performance of both models against SPY index benchmark in various real-world scenarios.